

COLD FUSION TIMES

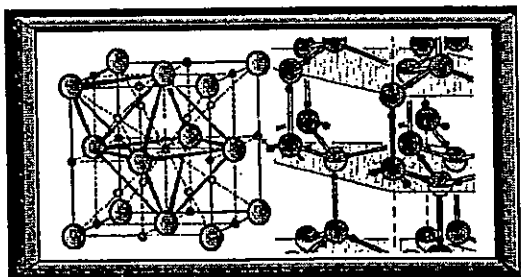
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THE JOURNAL DEVOTED TO
RELEVANT SCIENTIFIC AND
ENGINEERING RESEARCH OF
THE NEW HYDROGEN
ENERGY

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Metallic palladium (at left) is shown fully loaded with deuterium, an isotope of hydrogen. Heavy water (D_2O , on the right) is the source of the isotopic fuel used in some "cold fusion" reactions. Notice the hydrogen lattice within the palladium lattice.

4th International Cold Fusion Conference

from MAUI - Four days of idea swapping and resource sharing, intense lectures of theory, material science, nuclear physics, and a few developing businesses. The cold fusioners reported enjoying the "thinning out" of the few remaining skeptics along with having to deal with a cornucopia of many new papers and laboratories confirming cold fusion phenomena. New developments occurred. More than 150 papers, and of them more than 100 were experimental papers. Many are discussed here. So stay informed on Up-to-Date News from California, Boston, Illinois, Texas, Utah, as well as Russia, Italy, China, America, UK, Japan and France on CF materials, theories, nuclear products and emerging technology.

from China Lake, California

from Sapporo Japan

U.S. NAVY CONTINUES POSITIVE EFFORTS SUPPORTING COLD FUSION

At the International Conference on Cold Fusion (ICCF-4) there were scores of confirmations of nuclear and thermal products created by several cold fusion phenomena. These confirmations ranged in numerous systems including heavy water (with and without lithium), D_2 gas, and high voltage glow discharge systems. Many materials have now been successfully loaded with deuterium. The US Navy has distinguished itself by both generally supporting the field [including partially sponsoring the EPRI meeting] and by high caliber research including the continued efforts at demonstrating the linking of helium-4 with the excess heat. Melvin Miles and B. Bush, of the Chemistry Division, Research Department Naval Air Warfare Center Weapons Division China Lake, CA spoke of demonstrating "ash" consistent with a nuclear process, where that ash is helium-4. Their paper "HEAT AND HELIUM MEASUREMENTS IN DEUTERATED PALLADIUM" is the COLD FUSION TIMES' choice for the *Summa cum laude* experimental paper from a large laboratory given at ICCF-4. The following are excerpts from their important paper.

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RHODIUM: NEW MATERIAL DEVELOPMENTS

Much more information has appeared confirming rhodium as a useful alloy with palladium to be used as the material into which to load the isotope of hydrogen. Several papers appeared from several Japanese investigators at IMRA (Japan) including Drs. Akita, Hawegawa, Hayakawa, and Tshuchida. The following excerpts from one of the papers compares the Pd-Rh alloys to the previous much touted Pd-Ag alloys. This is followed by reviews from other labs of data involving silicon, thiourea, samarium, cerium, yttrium and more.

Continued on page 7



Prof. Dennis Cravens

Who is the choice of both the COLD FUSION TIMES' and Dr. Fleischmann's for author of the BEST ICCF-4 Experimental Paper? The COLD FUSION TIMES, based on Delphi-type polling around the United States, awards Dennis Cravens of Texas, for his exceptional personal achievements contributing towards an understanding of cold fusion, the Andrew M. Riley Memorial Citation for the Best Experimental Paper presented at the ICCF-4 International Conference.

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U.S. NAVY: PRODUCTS OF COLD FUSION HELIUM-4 IS LINKED TO THE EXCESS HEAT

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"Our previous results present a correlation between the measured excess power and helium production in D₂O-LiOD electrolysis cells using palladium cathodes. The measured rate of ⁴He production (10¹¹-10¹² ⁴He/s*W) is the correct magnitude for typical deuteron fusion reactions that yield helium as a product. ***** Metal flasks were used to collect the electrolysis gas samples in order to minimize atmospheric contamination due to helium diffusion through glass. The helium concentrations in Table II support a detection limit of approximately 10¹³ ⁴He/500 mL in these experiments as reported previously. Mean values for the measured helium concentrations in these control experiments are 4.4 +/-0.6 ppb or 5.1 +/-0.7 x 10¹³ ⁴He/500 mL. ... For experiments producing excess power, five helium measurements using these same metal flasks have been completed. These experiments yield a mean value of 2.0 +/-0.5 x 10¹¹ ⁴He/s*W after correcting for background levels of helium measured in control studies (Table II). This value is once again the correct magnitude for typical deuteron fusion reactions that yield ⁴He as a product."

from Texas:

PHOTON IRRADIATION INDUCES COLD FUSION

Photon irradiation activates cold fusion. This has been previously suggested and is now demonstrated using radiofrequency photons at the University of Texas (A&M) by Prof. J. Bockris (at ICCF-4, December 1993)]. Prof. Bockris discussed the triggering of the phenomenon by this method.

"Two types of triggering the Cold Fusion effects have been studied. The first type concerns electrochemical triggering, employing Low-High Current sequences of the type introduced by Bockris and Hodko and later by Takahashi. During this pulsing, the D/Pd ratio remained at about 0.83. The increase in the electrolyte temperature during the period of high current indicated an excess heat which was ~17% higher than that computed on the basis of classical electrolytic theory. The overpotential measurements on the Pd indicated a value of about 350 and 950 mV during low and high currents, respectively. The other kind of triggering involved Radio Frequency stimulation at MHz frequencies and at 20-100 mW intensity while maintaining the applied d.c. The temperature of the cell electrolyte increased within 30 minutes and the, excess heat,, amounted to approximately 24% above that expected on classical grounds."

[from "TRIGGERING AND STRUCTURAL CHANGES IN COLD FUSION ELECTRODES" J. O'M. Bockris, R. Sundaresan, D. Letts and Z. S. Minevski Department of Chemistry, Texas A&M University, Texas]

Corroborating their excellent work are many other results at ICCF-4 and elsewhere. We thank the U.S. Navy in supporting their efforts which have been affirmed elsewhere. ICCF-4 continued the demonstrations of science in this field, and was consistent with the results of the Third International Cold Fusion Conference (ICCF-3) reviewed by Dr. V. Rehn, United States Navy.

While the excess power-density levels approaching 1 kW/cm³ which, were reported in 1989, have not been reproduced or quantitatively verified, power levels of order 0.1-10 W/cm³ have been reported by many researchers using a number of different experimental arrangements and various substances. *** Perhaps the clearest scientific fact, at this time, is the hardest for physicists to accept: nuclear reactions apparently do occur in deuterium-loaded Pd, Ti, and probably in other solids."

[Rehn, V., Ahmad, I., "The Third International Conference on Cold Fusion"
Scientific Information Bulletin, Office of Naval Research Asian Office
NAVSO P-3580, Vol. 18, Jan. 1993]]

from Palo Alto, California

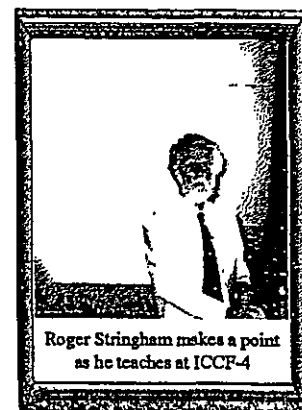
SONIC INDUCED COLD FUSION

Russ George and Roger Stringham left the audience wide-eyed with their presentation showing time-lapse imaging of collapsing bubbles filled with D₂. The bubbles were created by ultrasound using cavitation techniques. As the collapse of the bubbles occurs, the bubbles inject their gaseous contents into the metallic palladium. Helium-4 is formed and in some cases so much excess heat has been generated that the palladium has melted! Here is an excerpt.

"Cavitation Induced Micro-Fusion"

"Bubbles produced by an acoustic field adiabatically collapse creating very high energy systems called jets. When the liquid for the bubble formation was heavy water (D₂O) and the surface on which the bubble collapse was palladium foil, three phenomena were measured originating in the palladium foil lattice. The first was excess heat, where there was production of enough pulse heat to melt the lattice and enough continuous heat to sustain excess output of 90 watts for periods of up to three days. The second was the production of a small amount of ⁴He."

[photo: Roger Stringham E-Quest Sciences Palo Alto, California]



Roger Stringham makes a point
as he teaches at ICCF-4

Because of rapidly developing technologies in cold fusion research, the COLD FUSION TIMES remains available for home or office delivery.

COLD FUSION TIMES brings up-to-date, relevant information to its readers, and an open forum for all views on cold fusion.

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Questions, comments, suggestions, and contributors are welcome.

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POWER PROFILE OF THE MONTH

AT THE HELM OF EPRI --
DR. THOMAS O. PASSELL

As cold fusion scientists diligently continue to fine tune and confirm their research, there are certain people who herald special recognition for persevering in a storm of doubters. Sharing information and organizing ways for others in the field of cold fusion to better understand this phenomenon takes major league effort and commitment.

Thomas O. Passell, a Project Manager for the past 18 years at the Electrical Power Research Institute (EPRI) in Palo Alto, California, is this issue's Person of the Month for his steady support of cold fusion and by assembling and executing an outstanding Fourth Annual Cold Fusion Conference in Maui, Hawaii in December, 1993. The event



approximately 260 scientists and other cold fusion watchers at the Hyatt Regency for four days and nights of concentrated research and developments in the field.

Dr. Passell earned a Ph.D. in chemistry from the University of California in Berkeley, studying heavy elements above radium and electron spectroscopy. Most of Dr. Passell's past work at EPRI had to do with corrosion control in nuclear power plants in the steam making loop. Later, his primary interest and extensive experience focused on hydrogen and palladium at a similar project at SRI (Stanford Research Institute).

Dr. Passell's involvement in cold fusion was triggered by the famous Pons and Fleischmann announcement on March 23, 1989. On March 24, Dr. Passell wrote a memo to his group at EPRI who was in charge of initiating research. He said that the ease at which cold fusion could be tested without a very big investment was worth a try. "The electrochemistry is the nature of corrosion. And I had been in the corrosion field for the past 14 years when you get right down to it. I thought we ought to be able to get into this business from the ground floor. I thought 'if it's true than we're into something very big and if not, we would learn something new about palladium and hydrogen since we're interested in that for other reason's" said Dr. Passell.

A number of his colleagues felt cold fusion was "too good to be true". Based upon the flow of positive results, Dr. Passell continues to endorse and boost cold fusion research and discussions whenever he can. Nearly five years since the excess heat announcement, Dr. Passell, along with countless other scientists, believes the data speaks for itself. "I'll still feel good about it as long as there is enough support to do something." After all, he says, "I'm betting my career on it."

- by Gayle Verner

from Vernon, Texas

Continued from page 1

DENNIS CRAVENS

WINNER - 1994 ANDREW RILEY MEMORIAL CITATION

We congratulate Dr. Dennis Cravens of Vernon, Texas as the winner of the 1994 ANDREW M. RILEY MEMORIAL CITATION - for excellence in experimental work. This citation was suggested and originally published in Fusion Facts. Here are some excerpts from Dennis Cravens' abstract and paper presented at ICCF-4.

More in the next issue.

FACTORS EFFECTING THE SUCCESS RATE OF HEAT
GENERATION IN CF CELLS

Some factors have been empirically discovered to effect the success rate of heat generation in open CF cells. These include

PREVENTING CRACKING OF THE HOST LATTICE by using special alloys (10%Ag in Pd, 5% Re in Pd, 10% Mg in Ni, 15% La in Ni, 5% V/Sn in Ti), loading at low current densities ($< 60 \text{ mA/cm}^2$) until well into the beta phase, delaying addition of poisons (Al, Si, B, Thiourea) until after the beta phase, pre-loading by gas at elevated temperatures, adding materials to increase diffusion rates of D (Li) and using thin films of Pd on Ag to decrease loading stresses.

INITIATING REACTIONS by dynamic conditions by rapidly changing the temperature (changes of 30 degrees or heating to 80 degrees C), suddenly increasing the current density (by about an order of magnitude), exposing the cathode to B field variations in the RF range, quickly changing the pressure, vibrating the cathode with ultra-sound.

AVOIDING SOME KINDS OF CONTAMINATION by electrolytically cleaning the anode material, placing anode connections well out of the cell, keeping hygroscopic uptake to a minimum, and preventing inclusion of diamagnetic materials at the surface of the cathode.

SELECTING HOST LATTICE MATERIALS by avoidance of voids and fold formed Pd, uniform polishing of cathode surface, avoidance of sharp edges and convoluted forms, identifying good regions for electrodes by bubble formation and surface coloration, and using casted and cold rolled materials.

LOADING THE CATHODE IN A UNIFORM MANNER by selecting geometry to avoid large variations in E fields over the surface of the cathode, assuring the uniform surface texture of the cathode, avoiding sharp corners on the cathode, proper construction of anode, and using low current densities with high cell resistance ($> 5 \text{ ohms}$) during initial loading stages.

(Continued from Page 11)

News from the Internet - Hitting on DENNIS CRAVENS

* Dennis Cravens gave a very entertaining presentation on work done in his home lab. Lots of good slides but very few precise numbers were given. The talk was an attempt by Cravens to discuss the good "recipes" he had found for producing heat. Slow loading was recommended up to around .6 and then he polished the electrode with an aluminum oxide powder. He also used a finer photographic polisher afterwards. Then he resumed loading to higher levels by adding poisons to the electrolyte. Smooth cathode design and a large platinum anode covering most of the surface area of the cathode needed. You did not want any point to be able to outgas at a level greater than the rest of the Pd surface area. Showed the use of an 82 MHZ 100 milliwatt exciter a la Bockris. After loading when you are ready to go you need to suddenly raise the temperature and suddenly raise the current density.

In his closing remarks Cravens showed the use of a small cell inside the water tank of one of those small water pressure powered model rockets. He used lightwater cells and heavywater cells in these rockets. He then used boil off effects in the cells to launch. The lightwater cells were able to rise to an average height of 5cm. The laughing was very loud at this point but I believe the average height of a heavy water cell boil off was 30cm. Cravens said the patent office wanted to see the device actually doing something before granting any patent. Hmmm. It was a very well received talk and in his conference summary Dr. Fleischmann said that he enjoyed Dennis Cravens talk the most. I found myself clapping at the end. Cravens is the first guy that I have seen try to do anything with the alleged excess heat".

COLD FUSION THEORIES *****

This issue begins with a new column on theories of cold fusion. Edward Lewis offers both historic and physical theories and Harold Putoff has developed Zero Point Energy [ZPE] both of which are introduced here.

from Chicago, Illinois

The following is from Edward H. Lewis, Chicago, IL, a scientist with technical writing and superconductivity experience. He is developing theories involving anomalous physics phenomena and his hypothesis that anomalous phenomena are produced as sets at approximately 80 year intervals.

Luminous Tornadoes and Other Plasmoids

EDWARD LEWIS

Anomalous phenomena are produced as sets at approximately 80 year intervals. And new kinds of theories which resolved the phenomena were first formulated at approximately 80 year intervals since about 1506 when Copernicus first formulated his theory. The sets of phenomena have patterns, and one fundamental phenomena of the current set is that of plasmoid phenomena.

People have often seen bright or luminous tornadoes. According to prior research, a large percentage of tornadoes are bright or glowing, and people have experienced that some are quite hot(1). Storms on the Earth are probably an atmospheric manifestation of earth plasmoid activity, according to Tesla's experience of diminishing surges of electricity in the ground that accompanied a storm that was traveling away. Even clouds may be such a manifestation, a plasmoid phenomena. And clouds may convert to ball lightning. People have seen clouds which contained a glowing spot, and one case is fairly documented that a cloud with a glowing spot produced a tornado. Tornadoes seem to be a locus for the conversion of substance to light and electricity. The power of tornadoes is anomalously high. People have seen lightning from a large area converge to the area of a cyclone, but this seems to only be part of the reason for the power. People have seen tornadoes that had parts that were so bright that they described the phenomena as being too bright to look at though the tornadoes were quite a ways away. One person reported that as a tornado passed by it was "just as light as it is when the direct sun is shining" for about five minutes. One observer reported(2) that a tornado that had a part that seemed to be constantly bright passed 9 blocks away from him and was too bright to look at.

Tornadoes and ball lightning seem to be the same type of phenomena, though ball lightning is smaller. I would say that the cold fusion phenomena is a plasmoid phenomena. People have produced many types of phenomena including traces and holes and tunnels that are similar to those produced by plasmoid phenomena. It seems that tiny plasmoids like ball lightning are being produced. In several earlier papers I described that I suspected that much of the sparks that people see during their lives are actually tiny BL-type phenomena, and that these may be produced by fracturing rocks. The noise is probably similar to that which is produced by larger BL-type phenomena, except that it seems that larger BL-type phenomena produce lower frequency sounds perhaps together with higher frequency sounds. As an example of the sound of an extremely large size of BL-type phenomena in the atmosphere of the earth, tornadoes have been reported to produce a fairly low pitched roar like that of a very long and heavy train that is being pulled by a locomotive, but people have also reported a loud sound like millions of bees associated with another tornado.

W. Bostick produced that which he called plasmoids by discharging through electrodes. Bostick wrote a paper that was titled "Plasmoids" that was published in SCIENTIFIC AMERICAN in 1957(3). He may have been the first to apply this term to this phenomena. In this paper, he had already began to tell others about his speculation that galaxies and the phenomena he produced were similar. He compared the shapes and the travel of these things. He also speculated a little about the identity of "particles." He shows pictures of different kinds of plasmoid shapes in the article and related these to different kinds of shapes of galaxies. Many people including Bostick, Alfven who is a physics Nobel Prize winner, Peratt(4) and Lerner(5) have developed similar astronomical theories that model the universe as plasmoids and that can be said to be derivations or summarizations of the experimental work of W. Bostick and others. It has become evident that atoms can be defined as plasmoids, especially as according to the phenomena produced by Ken Shoulders(6).

Continued next issue ---

1) B. Vonnegut and J. R. Weyer, "Luminous Phenomena Accompanying Tornadoes," WEATHERWISE, 19-2, pgs. 66-68 (Apr. 1966); B. Vonnegut and J. R. Weyer, "Luminous Phenomena in Nocturnal Tornadoes," SCIENCE, pgs. 1213-1220 (Sept. 1966).

2) F. Montgomery, "Some Observations on the Tornado at Blackwell, Oklahoma 25 May 1955," WEATHERWISE, 9 (no. 3), 97 (June, 1956).

3) W. Bostick, "Plasmoids," SCIENTIFIC AMERICAN,

197, 87 (October 1957).

4) A. L. Peratt, C.-G. Faldhammar, and N. Rynn, "Guest Editorial: Plasma Experiments in the Laboratory and in Space," IEEE TRANSACTIONS ON PLASMA SCIENCE, 20 (no. 6), 573 (December 1992).

5) Eric Lerner, THE BIG BANG NEVER HAPPENED, New York, 1991.

6) K. Shoulders, REnergy Conversion Using High Charge Density, US Patent Number 5,123,039.

from Austin, Texas

THE ENERGETIC VACUUM: IMPLICATIONS FOR ENERGY RESEARCH H.E. Puthoff Institute for Advanced Studies at Austin

INTRODUCTION

Modern physical theory, specifically quantum electrodynamics (QED), tells us that the vacuum can no longer be considered a void. This is due to the fact that, even in the absence of matter, the vacuum is neither truly particle nor field free, but is the seat of virtual particle-pair (e.g. electron-positron) creation and annihilation processes, as well as zero-point-fluctuation (ZPF) of such fields as the vacuum electromagnetic field, which will be the focus of our study here. Formally, the energy density associated with the vacuum electromagnetic ZPF background is considered to be infinite. With appropriate high-frequency cutoffs the ZPF energy density is still conservatively estimated to be on the order of nuclear energy densities or greater. The enormity of the figures describing the vacuum electromagnetic zero-point energy raises the question as to whether these numbers should be taken seriously, whether they are due to some defect or misinterpretation of the theory, whether the ZPF fields ought to be considered as 'virtual' or 'real'. There is, however, no question but that the ZPF fields lead to real, measurable physical consequences. One example is the very real Casimir force, an experimentally-verified ZPF-induced attractive quantum force between closely-spaced metal or dielectric plates. An elegant analysis by Milonni, et al., at Los Alamos National Laboratory shows that the Casimir force is due to radiation pressure from the background electromagnetic zero-point energy which has become unbalanced due to the presence of the plates, and which results in the plates being pushed together. (We will discuss this effect in more detail later when we address the possibility of ZPF energy extraction.) Other effects which can be traced back to interactions involving the ZPF fields in a fundamental way include the Lamb shift (the slight perturbation of the emission lines seen from transitions between atomic states), the van der Waals chemical binding forces, the stabilization of atomic structure against radiative collapse, quantum field mechanisms underlying the gravitational interaction, and spontaneous emission.

ZERO-POINT ENERGY

To understand just what the significance of zero-point energy is, let us begin with a simple harmonic oscillator. According to classical theory, such a harmonic oscillator, once excited but with excitation removed, will come to rest (because of friction losses). In quantum theory, however, this is not the case. Instead, such an oscillator will always retain a finite amount of 'jiggle'. The average energy (kinetic plus potential) associated with this residuum of motion, the so-called zero-point energy, is given by: $\langle E \rangle = \hbar\omega/2$, where 'h' is Planck's constant

($h = 1.054 \times 10^{-34}$ joule/sec) and ' ω ' is the frequency of oscillation. The meaning of the adjective 'zero-point' is that such motion exists even at a temperature of absolute zero where no thermal agitation effects remain. Similarly, if a cavity electromagnetic mode is excited and then left to decay, as shown in Figure 2, the field energy dies away, again to a minimum value $\langle E \rangle = \hbar\omega/2$ (half a photon's worth), indicating that fields as well as mechanical systems are subject to zero-point fluctuations. It is the presence of such ZPF 'noise' that can never be gotten rid of, no matter how perfect the technology, that sets a lower limit on the detectability of electromagnetic signals.

If we now consider the universe as a whole as constituting a giant cavity, then we approach a continuum of possible modes (frequencies, directions) of propagation of electromagnetic waves. Again, even in the absence of overt excitation, quantum theory has us assign an $\langle E \rangle = \hbar\omega/2$ to each mode. Multiplication of this energy by a density of modes factor then yields an expression for the spectral energy density that characterizes the vacuum electromagnetic zero-point energy

$$\rho(\omega)d\omega = [\omega^2/\pi^2c^3]/[\hbar\omega/2]d\omega = (\hbar\omega^3)/(2\pi^2c^3)d\omega \text{ joules/m}^3$$

There are a number of properties of the zero-point energy distribution given in equation 1 that are worthy of note. First, the frequency behavior is seen to diverge as ω^3 . In the absence of a high-frequency cutoff this would imply an infinite energy density. (This is the source of such statements regarding a purely formal theory.) As discussed by Feynman and Hibbs, however, we have no evidence that QED remains valid at asymptotically high frequencies (vanishingly small wavelengths). Therefore, we are justified in assuming a high-frequency cutoff, and arguments based on the requirements of general relativity place this cutoff near the Planck frequency ($\sim 10^{33}$ cm). Even with this cutoff the mass-density equivalent of the vacuum ZPF fields is still on the order of 10^{94} g/cm³.

COLD FUSION TIMES

from California

from Cumberland, Maine

- 1993 Paxton Lectureship Winner

"Calorimetric Studies For Several Light Water Electrolytic Cells With Potassium Carbonate and Sodium Carbonate Electrolytes and Nickel Cathodes"

R. Bush and R. Eagleton

Physics Department, California State Polytechnic University*

and

Future Energy Applied Technology, Inc. (FEAT)**

Results will be reported for calorimetric studies with several cells employing light water based potassium carbonate and sodium carbonate electrolyte and employing nickel fibrex (fine nickel mesh) electrodes. Highlights will include the following:

(1) Two otherwise "identical" potassium carbonate electrolytic cells have been run in series except that the electrolyte of one employed de-deuterated water with only 1% of the heavy water contamination of the nanopure water of the other electrolyte. The fact that the excess powers at the same current density were identical within the limits of the experimental error supports the results of Bush's¹ earlier mixing experiments, which employed cells constructed by Eagleton, indicating that there is a genuine light water excess heat effect that is not due, as some critics have claimed, to the known heavy water contamination of light water.

(2) Certain impurities such as aluminum and silicon² appear to be important for achieving, and boosting, the heavy water excess heat effect. It would not be surprising that such co-factors may also exist for the light water excess heat effect. Experimental evidence is presented in support of a hypothesized co-factor for the light water excess heat effect achieved with a nickel cathode.

"NO COMMENT"

ASK MARILYN*

BY MARILYN VOS SAVANT



starting over—to

from
PARADE
Magazine
Dec. 19, 1993

SO, WHAT'S THE QUESTION?

The answer was "Cold Fusion." Here are some of your questions:

"What happened to Rudolph the Reindeer's tongue when he mistook the North Pole for a candy cane?"
—Leland Kuhn, Salisbury, Md.

"What was the tabloid headline when the two most famous Arctic explorers were married?"
—Don Shears, Overton, Nev.

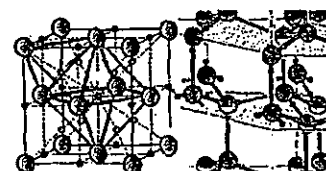
"If you have roast fusion for dinner one night, what do you have for lunch the next day?"
—John Geoghegan, Lexington, Ky.

The next answer is: "The Three Laws of Motion"

Marilyn vos Savant can be educated on cold fusion by writing ASK MARILYN, Parade, 750 Third Ave., New York, N.Y., 10017



Cold Fusion Impact
by Hal Fox, Ed. Fusion Facts
(see Page 10)



Charles Beaudette is a consulting engineer in Cumberland Maine. Trained at MIT, inventive in image compression, he now writes on political and technical issues including yellow rain and cold fusion.

COLD FUSION AND THE PRESS Charles Beaudette

"New developments in science have always held my attention. Beginning with my undergraduate days at MIT, I have followed them at a fairly technical level and I have continued to do so during 32 years of engineering practice. ... Although I am attracted to the technical story of cold fusion, my interest is also piqued by the public assault upon its two discoverers. Assault, by or through the press, upon a newly emerged public figure has become a deplorable commonplace of our communal life. The two discoverers of cold fusion now live and work in France and are financed by a Japanese company. It has come to that. The major media definitely agree with the Times on the meaning of the cold-fusion story. There is nothing there except unmitigated scientific hype. A number of refereed scientific journals, such as Fusion Technology, The Journal of Fusion Energy, The Journal of Electroanalytical Chemistry and several others are willing to run the gauntlet of elite opprobrium and publish cold fusion technical articles. The opposition to such articles, however, has not reached the point that it did during the anti-nuclear power days, when opposition to publication of particular articles, such as the Inhaber Report, was overtly organized and orchestrated. The preeminent scientific journals Nature, Technology Review, and Scientific American that write for a broad audience are now closed to technical news and articles on cold fusion. But more significantly, they are also closed to discussions about the merits of studying the excess-energy phenomenon. From their point of view, there is no scientific controversy about the existence of cold fusion. *** Editors of the scientific press are often the ones to whom the general press looks for the meaning of science stories. For example, the U.S. government's credibility in its 1992 charge that the Soviet Union was using a new T-2 mycotoxin-type of chemical weapon in Laos and in Afghanistan was effectively undermined by the appearance in Scientific American of a 10-page article purporting to show that "yellow rain," as it was called, was merely the excretions of honey bee swarms. A similar situation has arisen now with regard to the credibility of cold-fusion research results, but with a new twist.

As you may be aware, Scientific American is printed in many countries with more or less identical page content except for language. But in each country it is a separate publishing venture. The Nikkei publishing company publishes the Japanese version under the name Saizensu (Science). In the March 1992 issue of the Japanese edition, the results of Takahashi's experiments and consequent theoretical musings were well reported in a story covering two pages and with what I will call quantized information—what others might call actual numbers. Naturally enough, some American scientists called searchers periodically announce astonishing results. Akito Takahashi of Osaka University recently claimed excess power from an apparatus similar to the one originally used by Pons and Fleischmann. Yet Takahashi's results remain unreproduced by other researchers and unpublished in a peer-reviewed journal. Notice that when this was written it had only just happened. Twelve months after, Scientific American's readership had not yet been informed that Takahashi's work had been published in the International Journal of Applied Electromagnetics in Materials [3 (1993) 21] in 1992, nor had they been informed that his work had been reproduced by Dr. Edmund K. Storms at the Los Alamos National Laboratory, and published in Fusion Technology [23 (1993) 230] in 1993, even though editor Horgan admitted that Takahashi's claim represented "astonishing results." [Ed. Readers of the

COLD FUSION TIMES were aware] The critics chose to trust only the nuclear physics of 50 years ago. They pointedly ignored the law of energy conservation, a much older and more fundamental regimen. So what? There's nothing particularly wrong with being wrong, except being wrong. On such a matter as this, they will find themselves in excellent company, and a lot of it. But how are we to explain the largess of vitriol that has and is being slung at Stanley Pons and Martin Fleischmann? In this protracted conflict between theory and experiment always remember what it is that makes science science. Experiment prevails over theory. Theory never bows to experiment. The excess energy of the Pons and Fleischmann cell has been competently demonstrated. Finding its source is a wholly legitimate scientific enterprise now well underway. Nature and Scientific American magazines have greatly hindered this scientific work by heaping calumny on those scientists engaged in identifying its source. The scientific search for the origins of excess energy, and possibly immortal renown, will continue unabated in foreign laboratories. We must allow this exploration for the indefinite future as mother nature guards her secrets with great jealousy."

COLD FUSION TIMES

THE PATENT PAGE --

Analysis of the Intellectual Property Field in Cold Fusion

from the United Kingdom

First UK Cold Fusion Patent

The first patent involving cold fusion has been granted from the United Kingdom. A copy of this was graciously supplied by Mr. Gale Thorne (ENECO) and is presented here for the courtesy for our readers.



(12) UK Patent (11) GB (11) 2 231 195 (13) B

(54) Title of Invention

Thermal power generation by electrically controlled ion fusion

(51) INT CL: G21B 1/00

(21) Application No

9042218.4

(22) Date of filing

13.02.1990

(30) Priority Data

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(43) Application published

07.11.1990

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12.01.1993

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In accordance with Section 24(2) of the Patents Act, 1977, it is hereby certified that a patent having the specification No 2231195 has been granted to Harold Aapden, in respect of an invention disclosed in an application for that patent having a date of filing of 13 February 1990 being an invention for "Thermal power generation by electrically controlled ion fusion".

THERMAL POWER GENERATION by ELECTRICALLY CONTROLLED ION FUSION FIELD OF INVENTION

"This invention relates to the control of the fusion of hydrogen isotopes absorbed in a cathode conductor,

It is now known that, when a palladium cathode is used in the electrolytic decomposition of heavy water, the deuterons which find their way into the palladium can, once the concentration builds to a sufficient level, be caused to combine in what appears to be a process of nuclear fusion of the deuterons other than electrolysis.

BACKGROUND OF INVENTION

The basic problem concerning this fusion discovery is that of understanding how energy sufficient to overcome the fusion threshold can be available in low energy apparatus. An equally basic problem is that of understanding how the actions in a solid conductor can influence how two

deuterons combine there is the question of understanding whether hydrogen or deuterium absorbed into a metal retains its atomic form, with a satellite electron screening it from external matter. If the atom can shed its electron to contribute it to the free conduction electron population, then the hydrogen nucleus, be it proton, deuteron or triton is exposed.

In approaching these questions the inventor stresses that it is manifestly wrong to regard the nucleus of any atom as comprising neutrons and protons. Neutrons are only known experimentally in their short-lived free state and even then can be said to be formed from an antiproton and a fluctuating entourage of electron-positron pairs plus a positron. This is justified from the related explanation of single neutron diffraction and the precise theoretical derivation of the negative magnetic moment of the neutron. See the inventor's papers: 'A causal theory for neutron diffraction', Physics Letters A, vol. 119, p. 105 (1988) and, 'The theoretical nature of the neutron and the deuteron', Hadronic Journal, vol. 9, p. 129 (1988). Atomic nuclei, therefore, are undoubtedly composed of protons and/or antiprotons bound together by other charged particles, including electrons. The deuteron, in its most simple form, comprises two protons bound together by an intermediate electron. This is shown in the second of these references.

On this basis there is justification for regarding electrons as providing the binding action which can hold three protons together to form the triton from two deuterons which shed one proton in the process.

All that then is required to realize the possibility of deuteron fusion is an understanding of how enough energy can be available to stimulate two deuterons and an electron into a relationship close enough to allow the Coulomb forces to pull them together so that the repulsion of the positive charges is more than offset by the attraction of an electron sitting between them."

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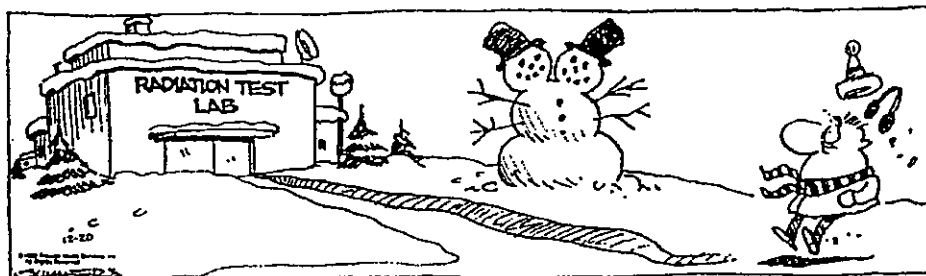
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Our Favorite Cartoon of the Month

by Dana Summers
"BOUND & GAGGED" THE
BOSTON
HERALD,
12/20/1993

BOUND & GAGGED

By Dana Summers



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MATERIALS CORNER — This issue: Rhodium, Samarium, Cerium and More

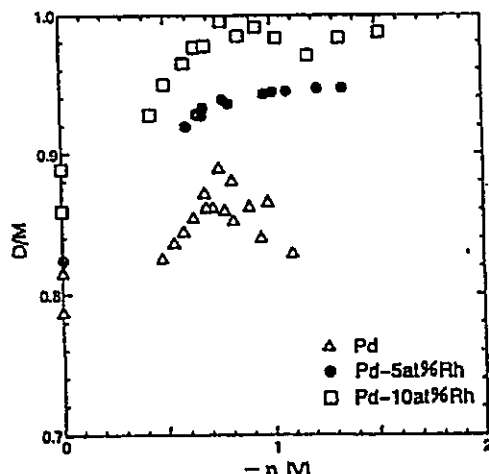
from Sapporo Japan

Continued from page 1

from Ekaterinburg, Russia

The following are excerpts from: "ELECTROLYTIC HYDROGEN / DEUTERIUM ABSORPTION INTO Pd, Rh AND Pd-Ag ALLOYS IN FUEL CELL TYPE CLOSED CELL" by H. Akita, Y. Tsuchida, Kubota, Kubota, Kobayashi, Yamamoto, Hasegawa, Hayakawa and Kunimatsu IMRA JAPAN CO. LTD.] The Rhodium story was

further investigated in comparison with silver by Hasegawa.



"The effect of cathode materials on the loading was studied by comparing the loading on Pd, Pd-Rh alloys and Pd-Ag alloys.

Cathode pretreatment was standardized to

polishing, degassing in vacuo at 200°C for three hours and electrochemical surface cleaning by scanning electrode potential between 0.4V and 1.5 V(vs. RHE) at 50 mV/sec. for ten minutes under hydrogen or deuterium gas before pressurizing the cell.

D/Pd on Pd higher than 0.9 can be achieved by surface modification."

D/M = 1.0 can be achieved for Pd-Rh alloys where $M = \text{Pd} + \text{Rh}$."

"Electrolytic Deuterium Absorption by Pd-Rh Alloy Cathodes in Fuel-Cell Type Closed Cell"; Nana HAYAKAWA, N HASEGAWA, Y YAMAMOTO, K KUNIMATSU [IMRA]

"The loading ratio on Pd-Rh alloys increases with Rh content as well as overvoltage. Maximum deuterium loading ratio on each cathode is as follows: D/Pd=0.89 on Pd, D/M=0.95 on Pd-5at%Rh, and D/M=1.0 on Pd-10at%Rh.

THE INFLUENCE OF CONDUCTIVITY ON THE NEUTRON GENERATION PROCESS IN PROTON CONDUCTING SOLID ELECTROLYTES

A. L. Samgin, A. N. Baraboshkin, I. V. Munin,
S. A. Tsvetkov, V. S. Andraev, G. Vakarin

620219 Institute of High-Temperature Electrochemistry,
Russian Academy of Science, Ekaterinburg, RUSSIA

The cold fusion process accompanying a generated nuclear emission (neutron, for example) was observed in solids with different conductivity types: in metals¹, tungsten bronzes², segnetoelectric³, and proton conducting solid electrolytes⁴. In this case the investigation of the phase transition influence resulting in a change of conductivity in the neutron generation process is of great importance. The proton conducting solid electrolyte, possessing only protonic conductivity in the high temperature region, was selected as an object of our research. The method of this investigation is described in this volume. The sample's thermocycling after filling the cell with deuterium and electrolysis revealed a neutron emission exceeding the background on the level 4 σ during definite temperature intervals. When a high deuterium pressure was produced in a cell (up to 3 atm), the sample's partial reduction was observed, accompanied by the sample turning black. The sample reduction resulted in a sustained electronic conductivity with air atmosphere at room temperatures. The fact that the neutron generation process stopped after the sample reduction may relate particularly to mixed electronic-cationic to electronic conductivity passage.

The investigation of this deuterium-solid state system's property permits a new interpretation of anomalous nuclear-electron effects. The same behavior is in agreement with one of the main conditions of cold fusion — the formation during deuterium saturation in strongly nonequilibrium conditions of the local inhomogeneous zones with an abnormally increased deuterium atom concentration relative to the lattice atoms⁵, with a hypothesis of forming the regions with dielectrical properties in Pd⁶.

Here we may mention that the discovered effect is in agreement with the neutron emission observed in monocrystals of tungsten bronzes² during the changing of the alkali metal concentration (for example, by replacing with hydrogen isotopes), which produces semiconductor and dielectrical layers. Current observations also correlate well with the investigations of deuterium emission from solids with segnetoelectric-dielectric or segnetoelectric-ion conduction⁷. This presents peculiarities of the nature of conductivity that may also be useful in understanding both mechanisms which have a common origin in electromagnetic current behavior in dense media, and capillary fusion in special structures with channels^{8,9}.

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from Denver, Colorado

NOVA: A COLD FUSION THERMAL ELECTROLYTIC CELL

"The Electrolytic Thermal Cell, or ETC, reflects the goal of assured safe operation at temperatures to 350C, pressures to 3500 psi (250 atm), and electrolysis input powers up to 1 kw. The core of an ETC unit is a high pressure electrolysis chamber of thickwalled stainless steel construction with a structurally embedded cast silver thermal dissipator in its lower section. The dissipator is tightly coupled to the calorimeter or boiler section, which has an integral, pressurized coolant (working fluid) reservoir ****

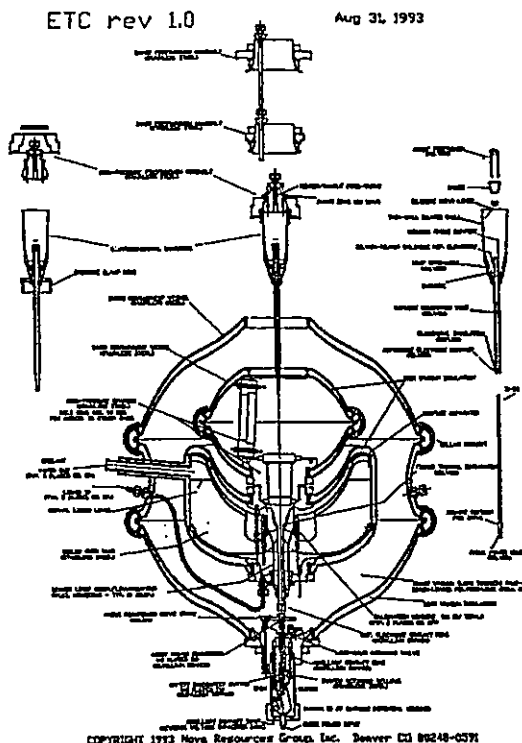
The containment vessels thermally isolate the experiment as well as protect the experimenter and laboratory from overpressure events. The outer containment vessel, which encloses the entire calorimeter and electrolysis chamber, and the inner containment vessel, which encloses only the upper portion of the electrolysis chamber, are vacuum insulated, double-wall, heavy gauge stainless steel shells, as are the calorimeter/boiler/reservoir walls. The normally evacuated space between the inner components and the outer vessel can also be filled with neutron moderation and absorption materials."

"The coolant, a fluorinated organic liquid, $\text{CF}_3\text{CH}_2\text{OH}\cdot\text{H}_2\text{O}$ is thermosiphoned from the reservoir past the thermal dissipator, where it boils.

Entrained droplets are then centrifugally separated from the vapor and returned to the reservoir. After a secondary demisting stage, the saturated hot vapor exits the ETC

to pass through flow measurement devices and into condensing coils. Recondensed coolant is pumped back into the boiler reservoir at its working pressure, up to 750 psi (50 atm)."

[from APPARATUS FOR SAFELY EXTENDING COLD FUSION INVESTIGATIONS TO HIGH Temperature, pressure, AND INPUT POWER REGIMES; H.E. Ransford, III, S.J. Pike Nova Resources Group, Inc. (ICCF-4; December 1993)]



from Massachusetts

JAHN-TELLER SYMMETRY BREAKING AND HYDROGEN ENERGY IN GAMMA-PdD:

Keith H. Johnson of M.I.T. Cambridge, MA presented his developing theory of cold fusion. The following is from his abstract discussing cold fusion as the storage of the latent heat of water.

In 1989 (*), we proposed a common quantum-chemical basis for superconductivity and anomalous electrochemical properties of palladium loaded with hydrogen and deuterium, derived from H-H/D-D bonding molecular orbitals at the Fermi energy between tetrahedral interstices ("gamma-phase" PdD). Symmetry-breaking anharmonic vibrations of the protons/deuterons, induced by the dynamic Jahn-Teller effect, promote superconductivity in PdH/PdD at $T_c = 9/10\text{degK}$, while the large vibronic anharmonicity explains the inverse H/D isotope shift of T_c . The calculated deuteron vibronic amplitude of 0.46A implies a closest D-D approach of 0.76A between neighboring tetrahedral sites and a fusion rate of only approximately $10^{**}(-24)$ per deuteron pair per second in gamma-PdD at ambient temperature, much too small to explain reported "excess"

heats. Ab initio quantum-chemical calculations for gamma-PdD further indicate that the "channels" connecting tetrahedral sites provide, via the Jahn-Teller effect, an "orbital pathway" for bulk catalytic recombination of atomic deuterium to rapidly diffusing dideuterium, the recombination heat equaling 9.4eV per Pd atom or 102KJ per cubic cm. of palladium per unit diffusion cycle time, equivalent to the storage and release of latent heat of 2.5 moles of heavy water. While the diffusion cycle time depends on cell conditions, for cycles between 1 and 100 minutes, this process could generate 17 to 1700 watts per cubic cm. of stored latent heat in gamma-PdD, enough to explain the Pons-Fleischmann "heat-after-death" phenomenon. The inverse isotope effect implies a slower hydrogen recombination to dihydrogen and slower dihydrogen diffusion in gamma-PdH, leading to negligible latent heat power from Pd-based light-water cells. Nevertheless, this mechanism could explain reported heat generation from light-water cells using nickel cathodes, where recombination of atomic hydrogen to dihydrogen is a rapid (110) surface or near-surface phenomenon.

* K.H. Johnson and D.P. Clougherty, Mod. Phys. Lett. B, Vol. 3, p. 795 (1989)

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MIT IAP 1994 3rd Annual COLD FUSION SEMINAR

(**) Sat. January 22, 1994 9:00-4:30 PM
Room 2-190 8:30 AM (Rm 6-120 after 11 AM)
Massachusetts Institute of Technology,
Cambridge, Massachusetts

In this day of scientific lectures, discussions, and videos, we will focus on the technologic aspects of cold fusion including updates from the Fourth International Conference (ICCF-4). Emphasis will be on calorimetry, materials, nuclear products, and theories which may explain the findings.

(**) IAP is the Independent Activities period at MIT. Open only to members of the MIT community, MIT students and alumni/a/c.

- | | |
|---------------|---|
| B. Ramsey | 76. "Conditions for Bose Condensation" |
| L. Smalley | "Cold Fusion: A Scientific Update" |
| H. Post | "Developing a Practical Fusion" |
| P. Hagelstein | 77. "Jahn-Teller Effect & Latent Heat of Water" |
| D. Pines | 78. "Cold Fusion: History and present status" |
| S. Hahn | "Thermodynamic Properties of Water" |
| D. Renshaw | "Current Status of Cold Fusion" |
| C. Bunker | "Cold Fusion: A Scientific Update" |
| L. Smalley | 79. "Materials and Cold Fusion" |
| L. Smalley | 80. "Materials and Cold Fusion" |
- Open Discussions - various engineering, materials, BMD subjects



Luminescence and microsparks of titanium bolts in potassium carbonate solution- from "Searching the Energy of the Universe - Photographs of AC Electrolysis of Water", I. Yukinuma and T. Matsumoto presented at ICCF-4

COLD FUSION TIMES

from Japan

continued from page 7

HIGH TECHNOLOGY MATERIALS PAGE -- PROTON CONDUCTORS

Many new materials were reported at ICCF-4. Another of the most interesting was the SrCeO_3 material described by Mizano.

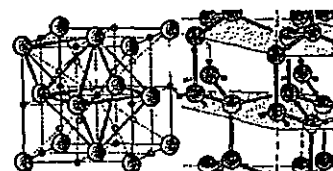
Anomalous Heat Evolution from SrCeO_3 -Type Proton Conductors during Absorption/Desorption of Deuterium in Alternate Electric Field by Tadahiko Mizuno, M. Enyo, T. Akimoto and K. Azumi, Hokkaido Univ., Sapporo, Japan

The cold fusion phenomena were tested with use of proton conductor solid electrolyte plates maintained at 300-400 deg C. An anomalous level of excess heat evolution of the order of 100 watt cm^{-2} was observed during absorption/desorption cycles of deuterium-containing hydrogen gas under application of an alternate electric field. *** A common system widely employed is the electrolysis of heavy water using Pd electrodes, in which one reason of experimental difficulties is that deuterium charging into the sample requires extremely long time. Takahashi (2) observed a large amount of heat generation by employing cyclic changes of current on Pd in heavy water electrolysis. This suggests that a kind of perturbation in electric current may be advantageous. Yamaguchi (3) reported a clear evidence of ^4He production from a Pd plate whose each faces were covered with MnO_2 and Au film, respectively, in this work, the sample was believed to contain comparatively low concentration of deuterium, and it was heated up to several hundred degree centigrade by an electric heater. This may mean that quick raise of temperature is advantageous, in spite of the view that solubility of deuterium is lower at elevated temperatures. However, the condition of heat and other reaction products showed very low reproducibility and no controllability. Common to these cases, a kind of triggering might be helpful in this reaction. With the consideration that the reaction may occur with a chaotic state caused by the movement of proton in an array of the atoms of the specimen, we have tried to use proton conductor ceramics at suitable temperatures and to apply an alternate electric field.

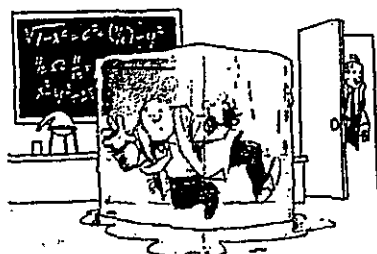
Samples were made from a mixture of SrCO_3 , $\text{CeO}_2\text{Y}_2\text{O}_3$ and $0\text{Nb}_2\text{O}_3$ powders. Mixed oxide powder was once sintered in a furnace at 1440 degC in air for 24 hrs, and was pulverized, again mixed and filtered to 400 mesh. The powder was pressed to form a plate of 0.8 cm diameter and 0.1 cm thick in a pressing machine and was again sintered in a furnace at 1460 degC in air for 24 hrs. Both sides of the sample plate were then coated with porous Pt film, 0.05-0.15 micro m, by applying Pt paste and calcination at 700 deg C for 24 hrs.

Sample was held with Pt plates of 1 mm thick pressed to both sides and fixed on a copper holder (5 x 4 x 2 cm) with a thin (0.1 mm) mica film in between Pt and a Ni plate which is connected with a stainless steel-coated thermocouple. ... Reaction cell is made of a 5 mm thick stainless steel cylinder with 8 cm diameter and 30 cm long. Sample temperature was kept constant by means of a stabilized power supply. The electric field was supplied from a function generator via. a power amplifier. The electric power was recorded continuously by means of a recorder and two digital voltmeters for current and voltage. ... The samples were heated by the heater and the temperature rose up to a constant level of e.g. 383 deg C. within one hr. The deuterium gas was then introduced into the reaction vessel: Temperature once fell down to 360 deg C, but it again started to rise, attaining 410 deg C after 5 hrs and this temperature was maintained at least for 20 hrs at which the experiment was terminated. in the other cases of hydrogen gas with the same type of sample or dummy sample in deuterium-containing hydrogen the temperature fell down to about 270 deg C within one hr and stayed there afterwards. This type of temperature fall should well be accounted for as due to increase of heat loss caused by thermal conduction of hydrogen. A small difference recognized between and hydrogen is probably caused by a small difference in their thermal conductivity.

The heat generation from the proton conductor in the experiment of deuterium-containing hydrogen gas was estimated to be approximately 50 watt (~ 100 watt cm^{-2}) over 20 hrs, or ~ 3.6 MJ in total. The input power given to the sample was +18 V, ± 40 micro A, or 7.2×10^{-4} watt. Accordingly, the output-to-input power ratio was estimated to be as large as 7×10^4 .



9



"Hey, Myerson...How goes the cold fusion experiment?"

TO COLDLY GO
WHERE NO ONE
HAS GONE
BEFORE

(left) from Parade Magazine
9/12/1993 LAUGH PARADE
by B. Hoest and J. Reiner

Despite the cartoons,
progress rolls on as the first
four issues of the
COLD FUSION TIMES
demonstrate

Peter Hagelstein (right) and Hal Fox (left)
were featured speakers at ICCF-4 who also
contribute
to the
MIT IAP
Cold
Fusion
Seminar
Series.

See
page 8 for
more
details.



from Podolsk, Moscow Region, Russian Federation

GLOW DISCHARGE EXPERIMENTS CONFIRM COLD FUSION

Further confirmation for the existence of nuclear products being formed by cold fusion (i.e. loading of isotopic material into a material) comes from other systems of cold fusion phenomena which now include both gas-filling systems and gaseous glow discharge systems in addition to the aqueous electrolytic system first described in March 1989 (Fleischmann). For example, although much much later, Kucherov (1993) has confirmed the cold fusion processes in the gas glow discharge systems.

"The results of four years of experimental work on glow discharge in deuterium with cathodes made of palladium and other materials are presented. About 500 experiments were made.

Discharge current varied in 10-100mA and voltage in 100-500V range ***** maximal measured excessive heat was about 30W and about 10KJ, about ten times the heat that could be produced in chemical reactions with existing deuterium and up to five times the electric input.

Excessive heat was observed in about 50% out of 78 experiments in which calorimeter was used.

In different experiments heat production was not correlated with neutron or gamma fluxes, but in a separate experiments it was. ***** Neutron fluxes with intensity up to 10^7 n/S were observed by activation of silver foil ^3He -filled detectors and scintillation detectors. Neutron spectra showed neutron energies up to 17MeV with anomalous shift to high energies (five orders) relative to d-d reaction. Gamma-spectrometry showed low level radioactive isotopes formation. Together with half-life time measurements it allowed to identify some of the isotopes, such as Rh and Sr isotopes. Most of the lines (~100) are still unidentified. Non-background gamma-lines sometimes can be seen for few days. Most of the gamma-lines appear in lower than 300KeV region.

X-ray films outside the chamber with led screens show some beam-like spots with energy 100-200 KeV.

Charged particles registration with SSB and CR-39 detectors showed good correlation of the results obtained by these methods. Maximal observed fluxes of charged particles were $\sim 10^6 \text{ S}^{-1}$.

X-ray film with lead screens showed X-ray fluxes up to 10^9 s^{-1} with soft (< 1KeV) and hard (10-30KeV) components. Sometimes characteristic X-rays of palladium can be seen with Ge-Li detector.

[from "Calorimetric and Nuclear Products Measurements at Glow Discharge in Deuterium"; Yan KUCHEROV, Alexander KARABUT, Irina SAVVATIMOVA Scientific Industrial Association "Luch", Podolsk, Moscow Region, Russian Federation (1993)]

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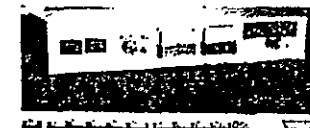
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NEWS FROM THE INTERNET

The view of one Conferee -- Mark Hittinger

[Ed: At Maui, all present reported that the cold fusioners were exceedingly gracious to the few skeptics who showed up to occasionally pitch questions which ranged from non-sequitur to irrelevant. The fusioners gave the skeptics time to speak and suggested offering a "FLYING PIG" award to the skeptic who contributes the best efforts to keep mainstream science away from the hard reality of cold fusion the longest.

Meanwhile on the nets immediately following ICCF-4, the fusioners have been silent. However, bugs@NETSYS.COM (Mark Hittinger) offered his comments. Some of his excerpts follow.]

"Random notes on ICCF4-Monday from an interested observer. Caveat: My background is Computers and Economics, not the natural sciences. ... In Japan they have

created "New Hydrogen Energy" with 30 million dollar funding over a four year period. NHE is underneath MITI.

Dr. Fleischmann gave a short talk on calorimetry with emphasis on "Simplicity and Accuracy". It looked good as this is one of the main problems with the current work. If you have never heard him speak I'll just say he is an old world smoothie and a pleasure to listen to. He began speaking of building a model of their calorimetry using linear and non-linear regressions....

McKubre - It looks like he is going to a lot of trouble to set up good experiments which will satisfy critics. One particularly nifty experiment involved a machine which would process a spool of palladium wire through several stages. Essentially the machine is a little "cathode adjustment" factory which then fed into a heavy water cell. Electrodes would reach out and touch the surface of the wire to induce electrolysis in the treated wire segment. I was worried that the wire segments were not "chopped" off before using (or didn't appear to be. The surface of the palladium wire could be treated in many different ways

before going into the cell to measure its excess heat if any. Yet another of a series of talks that seem to have an "optimization" theme to it. Instead of a mathematical model attempt McKubre looks like he is going to attempt to brute force a search for the more favorable conditions.

Bockris - speaks of the damage of cathodes induced by loading. A very interesting talk. If he loads too quickly the damage is greater. Damage allows D to escape easier. Damage prevents high loading ratios which are considered important to obtain the effect. Therefore damage and fast loading are bad. Load with low current density - under .35 A/cm² - this is considered good. Cathodes that produced excess heat were examined by surface leaching. Interesting "hexagon" shapes appear. Also "microvoids" little empty spheres around 2000 angstroms across. I was fascinated with an off topic discussion of using 82, 365, and 533 MHZ 100 milliwatt input to "excite/enhance" the reaction. The frequencies I list are just an approximation. Bockris showed math in his paper giving the actual frequencies to 4 decimal places, i.e. 81.xxxx Mhz. Others at the conference were very big on putting a coil of 100 milliwatt 82mhz energy around their cells. Bockris also says the effect is enhanced by the introduction of 200 gauss and 800 gauss magnets outside the cell. Another optimization kind of theme but one which does not involve the cathode impurity issue. Bockris appears to like near pure cathodes. [Ed. more on this in this issue - page 2].

Dr. Pons and "Heat After Death" - if you boil an open cell with light water until the water is gone the cell "dies". (The electrolysis stops) You then measure the temperature of the cell as the cathode cools. You get a cooling curve. If Dr. Pons does this with heavy water the temperature stays high and does not go down with the expected curve. The temperature will begin to drop off and then go back up a little before coming down again. It looks a lot like the curve of a crashing stock market - it doesn't go directly down and there are several attempts to move back up. The cooling curve of the heavy water cell is anomalous and P&F call this the "heat after death" effect.

Iwamura gave a detailed talk on loading Pd and then heating the Pd to release the D₂ gas. They did not try to achieve high loading ratios. It appears that they went to great effort to prevent background contamination. He presented a formula of (T production = DT ion content/total pressure). He stated that a near surface reaction is indicated, not a bulk effect. Using aluminum as a surface treatment yielded more neutrons and heat. I noticed that many people were playing around with aluminum this year. Claimed that T was more reproducible than neutrons or heat. Iwamura did not see any H/Pd results. The paper itself talks about T being 6 times the background but he also said that they had seen 40 times background.

Lida's group reported on foils of Pd and Ti. Aluminum surface treatment was mentioned. Unusual peaks of neutron emission at 3 and 7 Mev.

Srinivasan's group gave a talk on light water/tritium results. Many experiments were run with mostly nickel cathodes and light water. Excess heat is suppressed by the presence of anything made out of stainless steel in the electrolyte. On Monday Srinivasan stated that the Tritium production was seen in cells with Li electrolyte only.

Hittinger -- on Cravens Continued on Page 3

LETTER CORNER

NUCLEAR BYTES + HOT FUSION

* "PLEASE! You're ruining my eyesight.. ... Best,"
Arthur C Clarke, Sri Lanka

* "Just finished reading the #3 edition. You've got a good thing going there although the font is for younger people. Best wishes."
Charles Beaudette, Cumberland, Maine

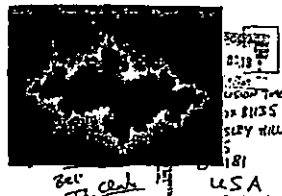
Sorry. We are progressively increasing in font size and will continue to do so as the support increases.

Thank you both for the note and careful attention.

* "I was very happy to that you are Cold Fusion Times as Hal Fox is Fusion Facts, George Miley is Fusion Technology I am convinced that (COLD FUSION TIMES) will play an important role in our war against ignorance, ill-will and stupidity.. A new journal is a joyful event. I wish you success and hope to contribute to this success!"

Peter Gluck, Institute of Isotopic and Molecular Technology, Cluj-Napoca Romania

Thank you very much for the support. George Miley and Hal Fox would probably also agree. ;-)



Dear Mr. Swartz
I am very happy to have that you are Cold Fusion Times as Hal Fox is Fusion Facts, George Miley is Fusion Technology I am convinced that (COLD FUSION TIMES) will play an important role in our war against ignorance, ill-will and stupidity.. A new journal is a joyful event. I wish you success and hope to contribute to this success!"
Peter Gluck, Institute of Isotopic and Molecular Technology, Cluj-Napoca Romania

from Rome

Nuclear Power in Homes

(Continued from Page 12)

present-day nuclear fission reactors on the scrapheap. In fission, atoms yield up their energy when they are split. In fusion, which powers the sun and hydrogen bombs, energy is released when atoms fuse together. But the process has yet to be taken beyond the realm of experimental reactors so expensive that countries have had to club together to afford them. Fleischmann and Pons aroused worldwide debate by saying their experiments showed nuclear fusion achieved with a simple apparatus made up of the metal palladium and deuterium, a form of hydrogen found in seawater. Pons said increasing numbers of scientists were now able to reproduce the energy-generating experiments, although they were still at a loss to explain exactly what was happening. "We stand by the fact that it is a nuclear reaction," said Pons, who runs the IMRA research institute in southern France.

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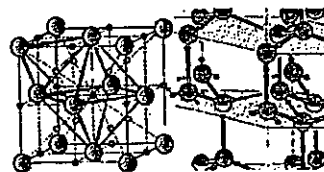
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TO COLDLY GO WHERE NO ONE
HAS GONE BEFORE

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NUCLEAR POWER IN HOMES POSSIBLE BY 2000 -SCIENTISTS

by Yann Tessier

Rome, Nov.28 (Reuters) - Two scientists who sparked a furor with claims of nuclear fusion in a test tube, promising unlimited, cheap energy, said on Sunday a household powerplant using the method was possible by the year 2000. Professors Martin Fleischmann and Stanley Pons triggered worldwide controversy in 1989 by saying they had harnessed fusion -- the energy of the stars -- with a simple apparatus costing a handful of dollars. Their findings astounded nuclear scientists who have spent billions of dollars trying to harness nuclear fusion. Scientists were unable to reproduce the experiments, leading to claims the two had made a mistake. But Pons told Reuters on Sunday: "The technology for a small device, enough for a household, is there." "I don't see why it couldn't be done in the next six years". ... The two have developed an apparatus which they say produces enough energy to keep a household light bulb alight for 15 minutes, but Fleischmann said they wanted to double its capacity every year. "To get a reasonably large device, one is talking of a few years, said Fleischmann, a professor at Southampton University in Britain and a member of the country's respected Royal Institute. He also spoke to Reuters by telephone. For over 40 years, physicists have pursued fusion as the answer to mankind's energy problems, a technology that would put

Continued Page 11

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